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Internship at Lightyear

Lightyear

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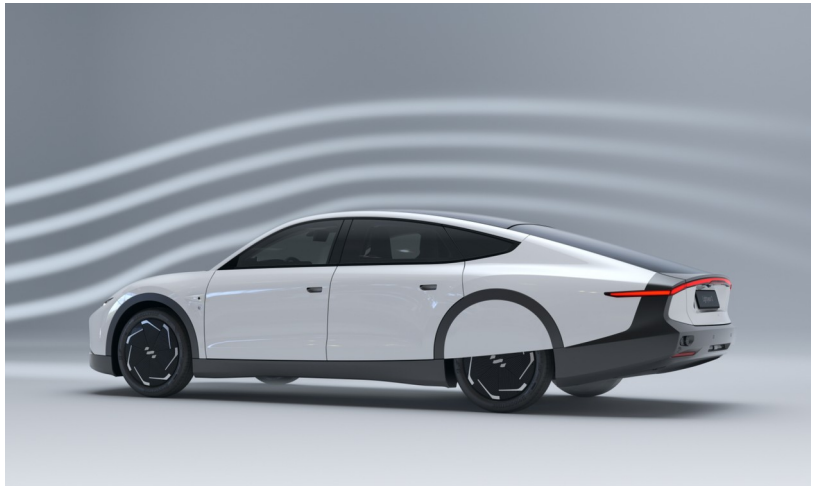
About Lightyear

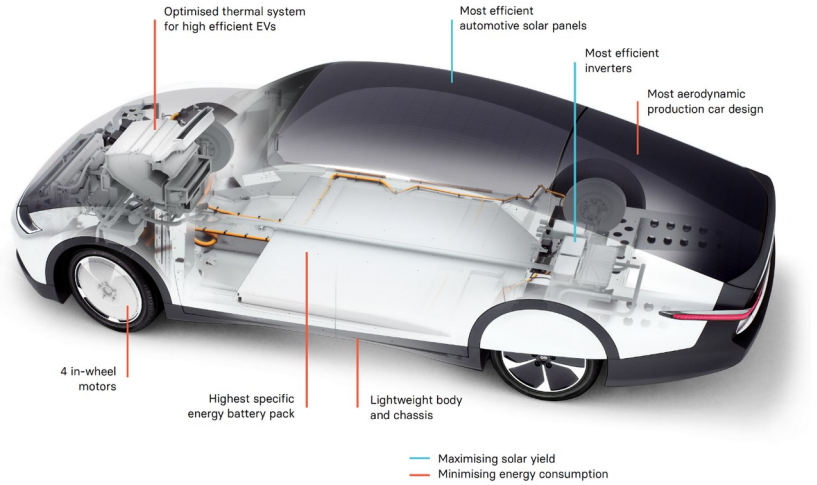
— Solar racing team TU/e

— Lightyear:

Providing clean mobility to everyone, everywhere.

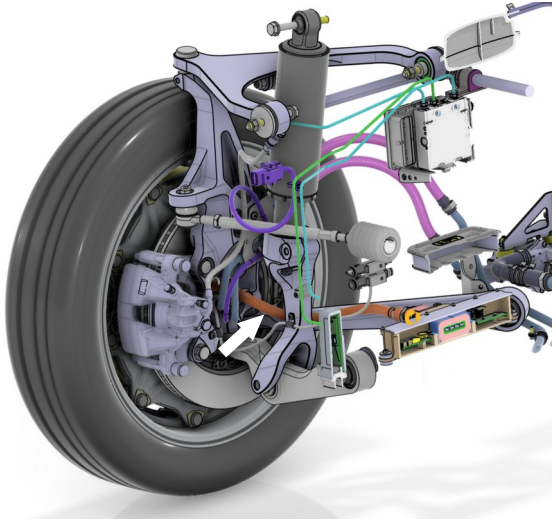
— Innovative technologies
  - Solar panels
  - Lightweight design
  - Most efficient inverters
  - Hub-motors

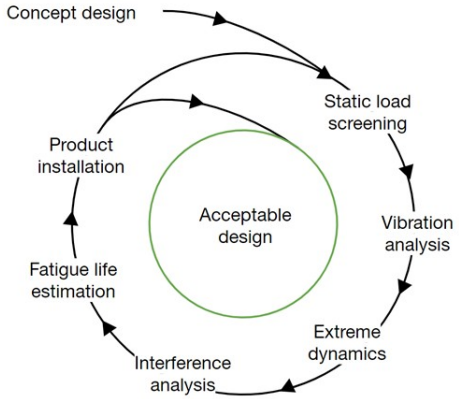




Internship assignment

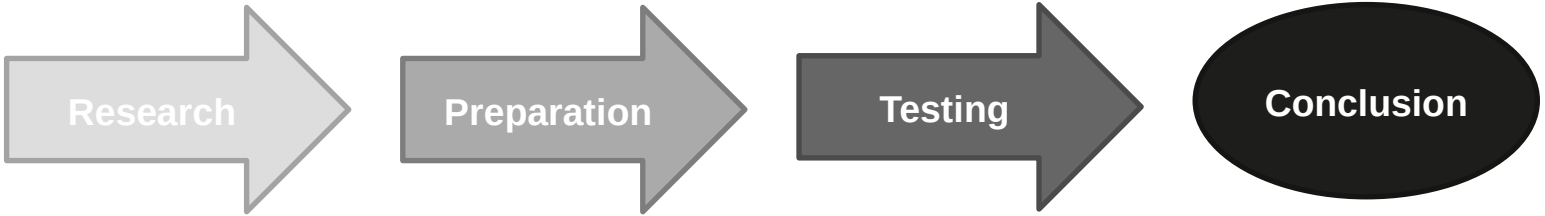
- 3-phase cable of the Lightyear 0
  - Damage concern
- Previous testing by Arne & Mats
  - Vibration testing in the cable
  - More testing required
- Life expectancy
  - Durability testing
- Dynamic loop
  - Test set-up





Timeline

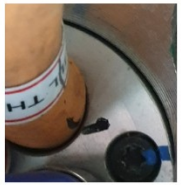
- Research and inspection
  - Damage concern
- Selection and preparation
  - Test set-up
- Durability testing and refining
  - Testing results
- Conclusion and recommendations
  - Advice on potential service interval

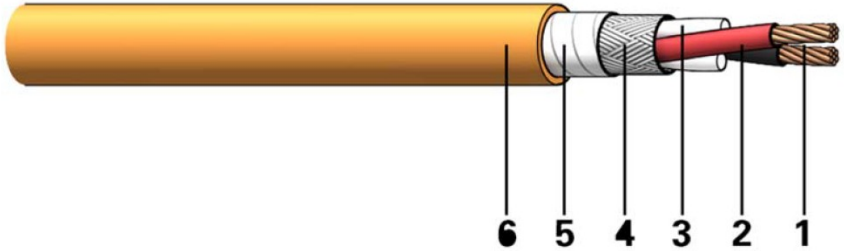


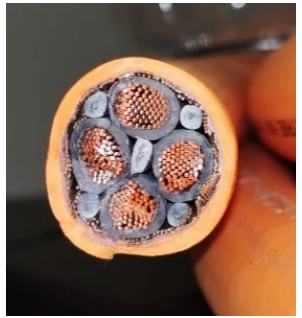
Vehicle inspection



- Damage concerns
  - Documentation of Sachin
- Movement of the cable
  - Extreme positions
- Clashing components
  - Coolant pipes, cables, clamps..

DV vehicle inspection table		Frequency (Either kms or instances(weekly) - based on vehicle test plan)		
		DV03	DV04	DV05
		20220810	20220810	20220810
FL	E - Bulging	Bulging observed near stator shaft end cap 	Minor Bulging observed 	Yes significant 



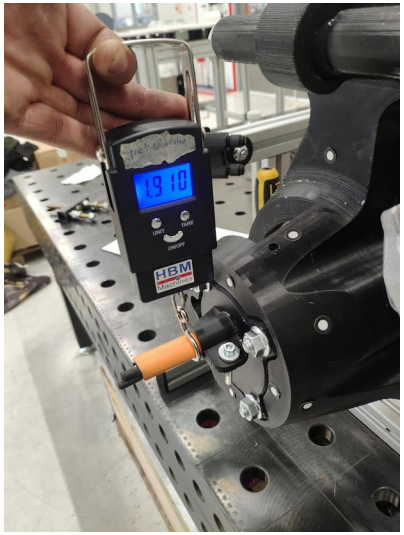


Terminology: Insulation, EMC-shielding,

Requirements for test set-up

- Bending force measurements
  - Test set-up of Arne & Mats



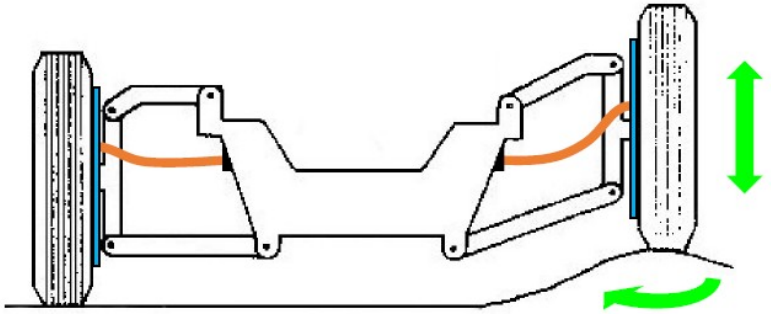


Lightyear

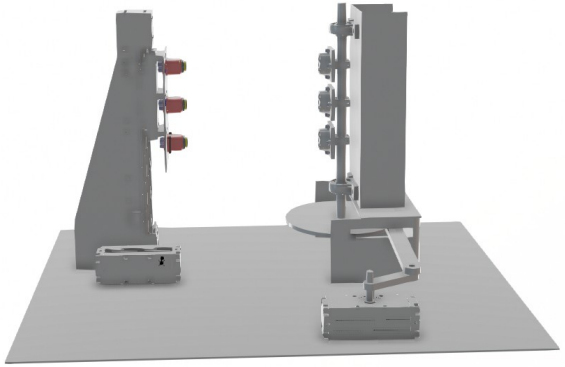
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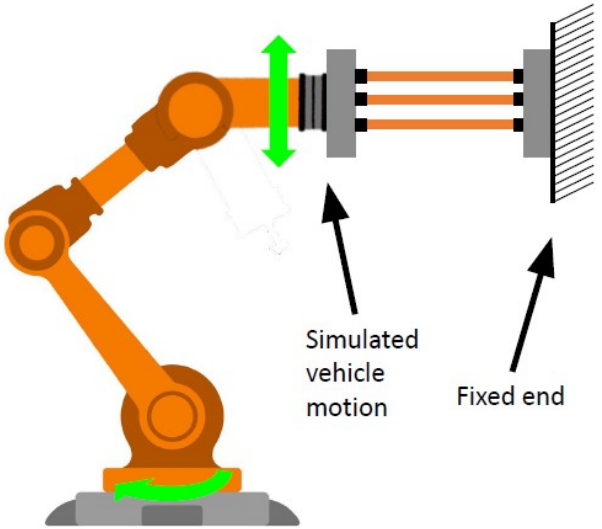
Test set-up options

- Custom build test set-up
  - Linear actuator(suspension travel)
  - Electric motor (steering motion)

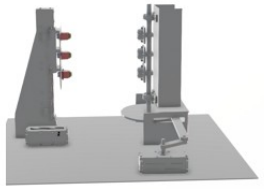


- Robotic arm
  - 4-axis





Choice table

Comparison Custom test set-up vs robotic Arm	 Custom test set-up	 Robotic arm
Costs Estimation based on testing	**5-6k** Mainly actuators Design, production, assembly (proto)	**15-24k** Denso SCARA 24k (full package) Roba & other suppliers contacted
Lead time Required preparation	**8-12 weeks** Design & ordering of custom parts	**3-4 weeks** Depends on delivery time
Flexibility Other applications	**Low** E-motor and linear actuator can be salvaged and reused	**High** Can be used for various other testing applications

Renting option:
3.8k per year

Preparation

- Preparation test location
  - Safety
- Program writing for robotic arm
- Planning
  - Start with bare cable
  - Continue with full cable assembly
- Cataloging necessary parts
  - Cables
  - Clashing components

```
Check:                                          'check set-up
          speed 5
          I[cycles]= 0
msgboxResult = MsgBox( "Checking set-up by running cycles at low speed", 0+64, "Set-
up Check" )

          I[test_cycles]= 0
                    move p, p[N]          'starting position
          do
                    call TestCheck
                              I[test_cycles] = I[test_cycles] +1
          loop until I[test_cycles]= 3
                    move p, p[N]          'starting position

          msgboxResult = MsgBox( "Executed " &I[test_cycles] &" cycles at low speed.
Start test?", 4+64, "Finished checking set-up" )
                     If msgboxresult = 6 then goto continue
                              'skip continue --> change to test
                     if msgboxresult = 7 then goto Check


Continue:
          I[velocity]= 10
          I[ramp_up_cycles] = 0
          do
                    speed I[velocity]
                    call Missionprofile
                              I[velocity] = I[velocity] + 15
                              I[ramp_up_cycles] = I[ramp_up_cycles] +1
          loop until I[velocity] = x
test:
          'move p, p[N]         '- starting position not needed
          I[startlogging] = 1
          speed x
          do
                    call Missionprofile
                              I[progress] = (I[cycles]/I[targetcycles])*1000
          loop until I[cycles] = I[targetcycles]
                    move p, p[N]
                     'starting position - finishing position minimal forces
givearm

          msgboxResult = MsgBox("Executed " &I[cycles] &"cycles", 0+64, "Finished")
                                        '1(lOK][Cancel]button) + 32(Warning Query)

End Sub
```



Lightyear

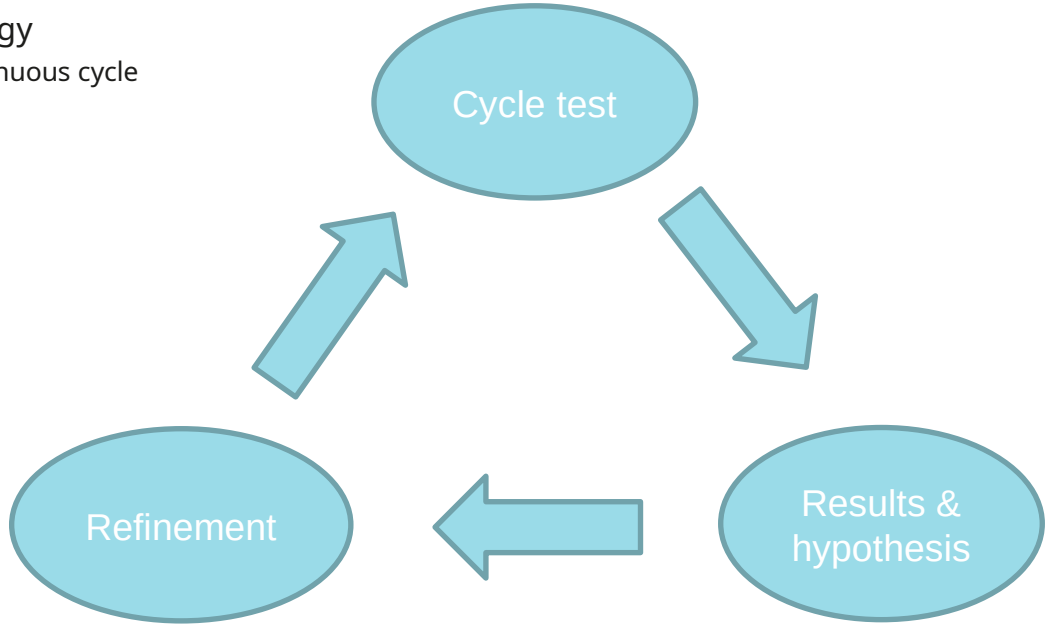
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Test set-up



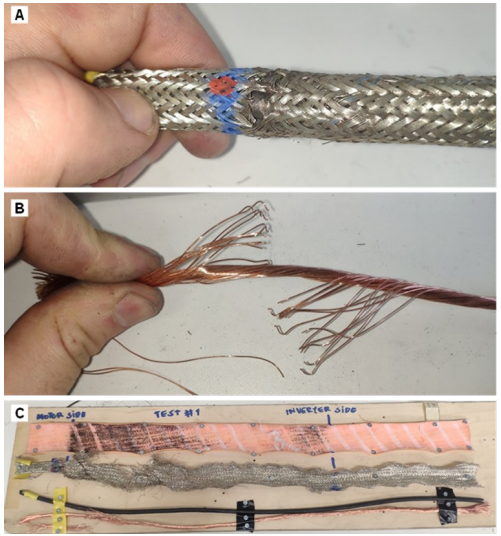
Testing

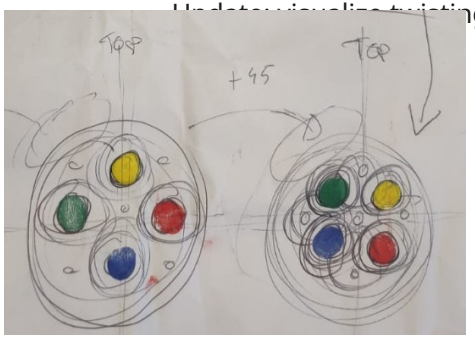
- Methodology
  - Continuous cycle

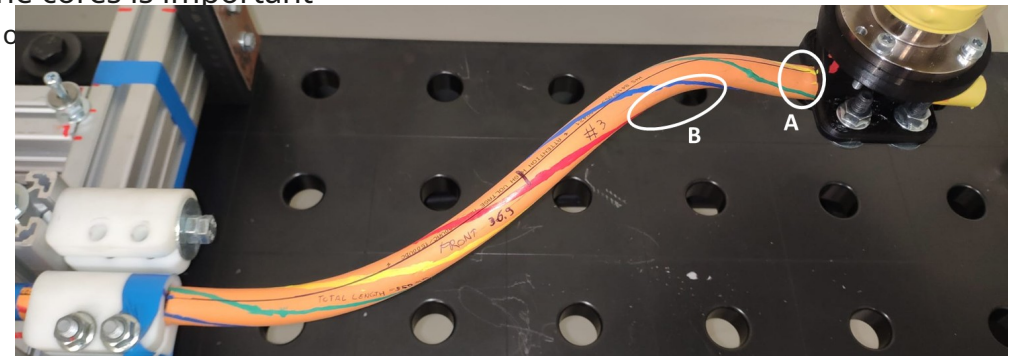


Example

- Test 1: 50k cycles
  - Broken strands: 15
- Test for datalogging: 150k
  - Broken strands: only 3
- Hypothesis: positioning of the cores is important





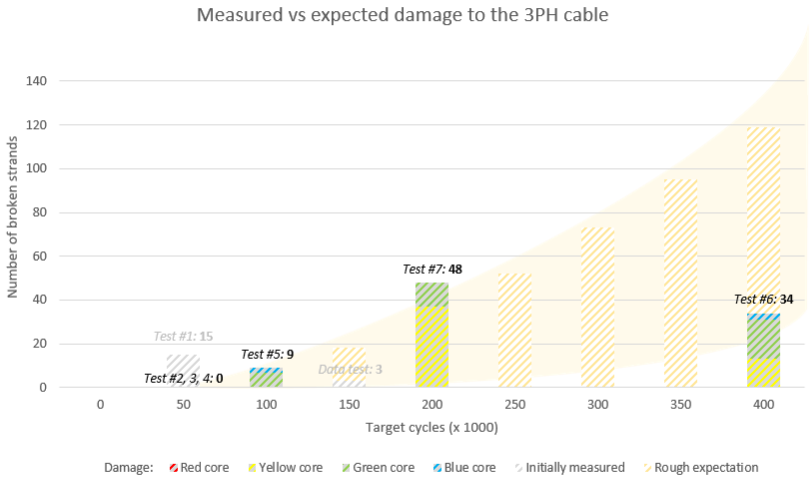


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Constraints within the assignment

- Fixed vehicle design and supplier agreements
  - No modification of the design
  - Possibility: proposing a life expectancy: service interval
- Limited supply
  - Reduced availability of parts from suppliers
- Bankruptcy of Lightyear
  - Lightyear 0 program scrapped
  - Abrupt end of the assignment
    - Incomplete results
  - LY2



Reflection

- Testing department
  - Could be a separate specialization at the HAN
- Learning new skills
  - Programming a robotic arm
- Importance of documentation
  - Reference for colleagues and future projects
- Experience bankruptcy
  - Without major personal consequences
- Amazing internship
  - Interesting assignment
  - Company goal: sustainable future
  - Great colleagues
  - Cool opportunities: IDIADA, Lightyear band, etc.

Lightyear

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Thank you.

Lightyear

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Hey Sammy,

Thanks so much for the work you did during your project at LY, but most of all thanks for being my travel buddy during the trip around the world to IDIADA! I'm sure you're going to do great in the future!

Mark

Sam mate, it's been real. Was great having you, thanks for bringing your high energy every day, I'm sure this will help you progress anywhere you go. All the best to you!

- Kris

Hi Sam,

Was really great having you in the physical team, you are a very ambitious and smart guy and enjoyable to have around. Always enjoyed coming over to see your progress, every week something new.

Wish you the best,
Jeroen

Hi Sam,

Thanks for your incredible effort over the past few months. Building your own test rig in a matter of months is an amazing achievement! I am sad that it has to end like this, but I am sure our paths will cross again at some point.

Arne

Hoi Sam,
Super bedankt voor al je hulp. Het was fijn om met je samen te werken en met jouw praktische insteek en goede communicatieve skills heb jij een mooie bijdrage geleverd aan het project.
Succes met je volgende uitdagingen en ik zou het fijn vinden om weer eens met je samen te werken Mats

Sam, wishing you all the best for future. Thanks for the efforts and contributions to the team!

Yogita

Hey Sam,

Thanks for all the work you did here at Lightyear!
Following your dream and I'm sure you do great in the future!

Ivo

Hey Sam!
I really enjoyed working with you! You have a great curious and quick mind. I was really looking forward to working with you further on your test setup, but helaas.
Wishing you all the best with your future projects!
Cheers,
Andrew

Ha Sam, ik zal die gekke dag niet snel vergeten dat 's nachts nog op en neer naar Amsterdam gegaan zijn met DV05! Super gezellige rit! Succes met de Renaultjes!

- Bram

Chiao Samuelle,
Thanks for spending all LY money with the robot and thus making the company bankrupt :)
Adios amigo

Nathachatte

Sam, you are a very positive person. With a really nice drive to go for it. The best bottle opener-programmer I have every met. I really liked and appreciated that you developed the first ever LY testing robot. Thanks for that. And keep up this spirit and go for it!
Kristof

Dear Sam, thanks for your hard work and commitment to make your project work as well as being a team player! Good luck with your next challenge! Hope to see you soon!
Chris Vorster

Team Lightyear



V2.0 220609